

Healthy Skepticism · Session 2 take-home summary

Fault 2: probability reversal

Session 2 is about one fault: probability reversal. A test reports a probability in one direction, and the question you actually care about runs in the other. The two sound the same and are not.

What can be measured is $P(\text{observation} \mid \text{condition})$: when the condition is there, how often do we see this? What you want to know is $P(\text{condition} \mid \text{observation})$: having seen it, how likely is the condition? The fault is reading one as if it were the other. They are different numbers, and how far apart they sit depends on something the observation never tells you: how common the condition is.

How to read a waffle chart

A waffle chart is a 10 by 10 grid: one square, one member of the population, one hundred squares in all. Give each member two attributes, A and B. Every square then lands in exactly one of four groups. In the classroom example (the numbers are illustrative): both A and B, 12% of the population; A only, 8%; B only, 20%; neither, 60%.

Writing $P(A \text{ and } B) = 12\%$ adds nothing mysterious. $P(\dots)$ is just a name for a percentage of the grid, and the four groups add to 100% because every square is counted once.

What $P(A \mid B)$ actually means

$P(A \mid B)$ reads "A given B." Because we are assuming B is true, the question becomes: of the B population, what percentage are A? Conditioning is a restriction of attention, not a shrinkage of the world. You set aside every square that is not B and count within what remains.

In the classroom example there are 32 B's in the grid (12 with A, 20 without) and 20 A's (12 with B, 8 without). So:

$$P(A \mid B) = 12 \text{ of } 32 = 37.5\% \quad P(B \mid A) = 12 \text{ of } 20 = 60\%$$

The same 12 squares sit in the middle of both calculations, and the two answers differ anyway, because the denominators differ. That is the whole engine of the reversal: flipping the conditional silently swaps one denominator for another.

Test results are conditional probabilities

A test result tells you which population you are now standing in. Let A be "the test says positive" and B be "the person actually has the condition." A worked example with illustrative numbers: the condition has a prevalence of 3%, the test has sensitivity 96% and specificity 95%, and 10,000 people are tested.

	Has condition	No condition	Total
Test positive	288	485	773
Test negative	12	9,215	9,227
Total	300	9,700	10,000

Read down the "Has condition" column for sensitivity: given the 300 who have it, $P(\text{positive} \mid \text{condition}) = 288 / 300 = 96\%$. That is a property of the test. Read across the "Test positive" row for what a patient asks: given the 773 who test positive, $P(\text{condition} \mid \text{positive}) = 288 / 773 = 37\%$.

More than half of all positive results in this example are false alarms: 485 of 773. That is why a positive result on a screen sends you for a second, more specific test, not to treatment.

Two reversals in the wild

A screening mammogram is sold on its sensitivity: $P(\text{test positive} \mid \text{has cancer})$ is high. But the reversed conditional, $P(\text{has cancer} \mid \text{test positive})$, is low, because the false positives outnumber the real cancers. A COVID rapid test is sold on its specificity: $P(\text{test negative} \mid \text{not infected})$ is high. But $P(\text{not infected} \mid \text{test negative})$ is not high enough, and the false negatives walk out infectious.

Both tests report $P(\text{result} \mid \text{condition})$. What you need is $P(\text{condition} \mid \text{result})$, and reversing it fails in opposite directions: the mammogram floods you with false positives, the rapid test sends you home with a false negative. The rest of the session walks each one with real numbers.

The first reversal: a screening mammogram

WHAT THE TEST IS MEASURED ON

A positive mammogram is a callback: the radiologist wants more pictures. It is not a diagnosis, and nothing has been found yet. The mammogram is judged on its sensitivity, $P(\text{called back} \mid \text{cancer})$, which is about 87%: of every 100 women with cancer, 87 are called back and 13 are missed. It is a good number. But notice where it starts: from already knowing she has cancer. That is the only direction a test can be measured in.

Sensitivity 86.9%. Lehman CD et al., National performance benchmarks for modern screening digital mammography, Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58. 1,682,504 screens, 359 radiologists, 95 facilities, 2007-2013.

WALKING THE REVERSAL

Apply the benchmark figures to an illustrative cohort of 10,000 women screened once. About six in a thousand have breast cancer, an average across ages 40 to 89 (it climbs with age: nearer 4 per 1,000 in the fifties, 6 or 7 in the sixties). The cohort splits four ways: 51 have cancer and are called back, 8 have cancer and are not, 1,103 are called back without cancer, and 8,838 are correctly cleared. The false-alarm group is more than twenty times the size of the group the test is looking for.

First question: of the 59 women who have cancer, how many get a callback? 51, so $P(\text{callback} \mid \text{cancer}) = 51 / 59 = 87\%$. Second question, the one people get backwards: of the 1,154 women called back, how many have cancer? 51, so:

$$P(\text{cancer} \mid \text{callback}) = 51 / 1,154 = 4.4\%$$

Twenty-two of every twenty-three callbacks are false alarms. This is not a trick of the arithmetic: 4.4% is the positive predictive value the benchmark study itself publishes.

FOR ONE WOMAN

87% is what the mammogram was measured on. 4% is what her letter means. Same test, same day, same woman. Unlikely is why she should not panic. Not ignorable is why the follow-up is warranted: among women recommended for biopsy, about one in four does have cancer.

Positive predictive value of an abnormal interpretation 4.4%, of a biopsy recommendation 25.6%. Lehman CD et al., Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58.

Why reasonable people disagree

Of US women aged 50 who are screened every year for ten years, published estimates say 0.03% to 0.32% avoid a breast cancer death, 49% to 67% have at least one false alarm, and 0.3% to 1.4% are overdiagnosed and treated for a cancer that would never have harmed them.

Nobody in this argument disputes the sensitivity, the specificity, or the 4.4%. The disagreement is over how to weigh a false alarm for half the women against a death averted for a fraction of one percent. So the guidance splits: the American College of Radiology recommends annual screening from 40, the US Preventive Services Task Force recommends every two years from 40, and the Nordic Cochrane group and the Swiss Medical Board have argued for not running screening programmes at all.

Ranges from published estimates for women aged 50 screened annually for a decade. Overdiagnosis is contested: the Canadian National Breast Screening Study's 25-year follow-up put 106 of 484 screen-detected cancers (21.9%) as overdiagnoses; Gotzsche and Jorgensen, Cochrane 2013 (CD001877), found no effect on overall mortality across ten trials.

The second reversal: a COVID rapid test

WHAT THE TEST IS MEASURED ON

The rapid swab is judged on its specificity, $P(\text{negative} \mid \text{no COVID})$, which is about 99%: of every 100 people who are well, 99 test negative. Almost nobody healthy is wrongly flagged, which is about as specific as any test in common use and is why a positive swab is worth believing. But look where that number starts: from already knowing the person is well. Nobody handed a negative result knows that.

Rapid antigen tests in symptomatic people: specificity 99.1%, sensitivity 73.0% (95% CI 69.3 to 76.4). Dinnes et al., Cochrane Database of Systematic Reviews 2022;7:CD013705.

WALKING THE REVERSAL

Take an illustrative cohort of 10,000 people, all with symptoms during a wave, so the odds before testing are far higher than in the street outside: four in ten have COVID (that pre-test figure is illustrative, chosen to represent symptomatic testing during a wave). With sensitivity 73% and specificity 99.1%, the cohort splits into 2,920 who have COVID and test positive, 1,080 who have COVID and test negative, 54 false positives, and 5,946 correct negatives.

There are 7,026 negative results, and 1,080 of them belong to people who have COVID. So:

$$P(\text{no COVID} \mid \text{negative}) = 5,946 / 7,026 = 85\%$$

99% is what the swab was measured on. 85% is what your negative means. More than one person in seven who was told they were negative had COVID and went out anyway. A negative narrows the odds. It does not clear you.

When nobody catches it: courtrooms and classrooms

The same flip, loose in the world, with nobody running the second calculation.

A RAPE TRIAL IN ENGLAND

The forensic expert told the court that the likelihood of the DNA sample coming from any man other than the accused was one in three million. That number describes how rare the profile is: $P(\text{match} \mid \text{innocent})$. It was offered as the chance he was innocent, $P(\text{innocent} \mid \text{match})$, the judge repeated it to the jury, and the Court of Appeal quashed the conviction in 1994. Lawyers call this the prosecutor's fallacy.

R v Deen, Court of Appeal, reported The Times, 10 January 1994; rules on presenting DNA statistics were later set in Doheny and Adams [1997] 1 Cr App R 369.

TRAFFIC STOPS IN HOUSTON, 2004 TO 2015

A two-dollar roadside kit turns blue for cocaine. It also turns blue for dozens of other things, which has been known since the 1970s. Blue was taken to mean cocaine. The lab checked only afterwards, and found that more than 250 people had already pleaded guilty to possessing nothing.

Houston Forensic Science Center and the Harris County conviction integrity unit, reported by ProPublica and The New York Times Magazine, 2016.

THE DETECTOR THAT CANNOT TELL

AI writing detectors are advertised on the forward conditional. Turnitin has claimed 98% accuracy, Copyleaks 99.1%, GPTZero 99%, Winston AI 99.98%. Every one of those is a statement about machines: if this text came from an AI, we will flag it. A flagged student needs the reverse: of the essays you flag, how many were actually written by an AI?

Vanderbilt ran that number. Even at a 1% false positive rate, their 75,000 submissions a year would falsely accuse 750 students, and they switched the tool off in August 2023. And 1% is generous: Stanford researchers ran seven detectors over 91 TOEFL essays, every one written by a human, and 61.3% were flagged as machine-written, with about one in five flagged by all seven. On essays by native English speakers the same detectors almost never erred.

All three stories end the same way: a number about how often this happens to the innocent was read as the chance of innocence.

Liang W, Yuksekgonul M, Mao Y, Wu E, Zou J. GPT detectors are biased against non-native English writers. Patterns 2023. Vanderbilt University disabled Turnitin's AI detection in August 2023. Vendor accuracy figures are the vendors' own marketing claims.

What to ask

When a test result is quoted at you, the number offered is almost always $P(\text{result} \mid \text{condition})$, because that is the only direction a test can be measured in. Before acting on it, ask: out of everyone who got this result, how many actually have it? That answer depends on how common the condition is, which the result alone never tells you. A positive on a screen is a reason for a better test. A negative narrows the odds without clearing you.

Next session: arbitrary categories

The next fault is arbitrary categories: a line drawn across a continuous measurement is read as a fact about your body. The question to bring: where do I sit on the curve, not which side of the line?

Materials and contact

Slides, take-home papers, and the whitepaper library are at healthy-skepticism.com.

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